

Fyn AI — Full Tool Audit

Generated 2026-05-01 from branch `fix/persona-split-review-fixes`. Source files audited:

- `app/Services/AI/AiToolDefinitions.php` (Anthropic-format schemas)
- `app/Services/AI/XaiToolDefinitions.php` (xAI/OpenAI strict-mode schemas)
- `app/Services/AI/AdviceFyn.php` (advice-mode tool stripping)
- `app/Agents/CoordinatingAgent.php` (`executeTool` dispatch + every handler)
- `app/Services/Onboarding/OnboardingChatDirector.php` (handoff path)

This is the complete catalogue of tools the Fyn AI ever runs. Every tool listed here corresponds to:

1. A schema definition (the JSON Schema the LLM sees).

2. A handler method in `CoordinatingAgent` (the PHP that actually runs).

Two-Fyn architecture (canonical contract)

The Fyn AI is split into two states behind one chat surface. The user never sees the switch.

State	Source	Mode	Tools available
Onboarding Fyn	<code>app/Services/Onboarding/OnboardingChatDirector.php</code>	The only state that enters or edits information	All capture/create/update/delete tools + the <code>delegate_to_capture</code> / <code>capture_complete</code> handoff stubs. Onboarding extraction tools (<code>capture_personal_details</code> , <code>capture_spouse_details</code> , <code>capture_dependants</code> , <code>capture_work_details</code>) are surfaced only via <code>toolsListOverride</code> during grouped-extract turns.
Advice Fyn	<code>app/Services/AI/AdviceFyn.php</code>	Read-only. Answers questions using engines/modules.	Read tools only. Every persistent record-creation tool is in <code>AdviceFyn::WRITE_TOOLS</code> and stripped from the catalogue. Even <code>navigate_to_page</code> is stripped (BS-14: the LLM was using navigation as an escape hatch for write intents).

Write intents in advice mode flow: LLM emits `delegate_to_capture` → `AdviceFyn::wrapStream` → `OnboardingChatDirector::handleInlineCapture` → the same direct-write handlers in `CoordinatingAgent`. The synthetic `handoff` SSE event is consumed internally and never reaches the frontend (INV-2.4.1).

Tool registration (where the tools come from)

`AiToolDefinitions::getTools(bool $isPreviewMode)` in `app/Services/AI/AiToolDefinitions.php:12` returns:

```
$tools = [
    ...$this->navigationTools(),    // navigate_to_page
    ...$this->analysisTools(),      // list_records, list_goals, list_life_events, get_module_analysis,
    get_recommendations, search_conversation_index
    ...$this->taxTools(),           // get_tax_information
    ...$this->planGenerationTools(), // generate_financial_plan
    ...$this->billingTools(),       // get_subscription_status, list_invoices, get_current_plan
];

if (! $isPreviewMode) {
    $tools = array_merge(
        $tools,
        $this->whatIfTools(),        // create_what_if_scenario
        $this->dataCreationTools(),  // creates: goal, life_event, savings, investment, holding, pension,
        property, mortgage, protection, asset, liability, gift, will, update_will, lpa, update_lpa
        $this->additionalCreationTools(), // creates: family_member, trust, business_interest, chattel
        $this->dataModificationTools(), // update_record, delete_record
        $this->profileTools(),         // update_profile
        $this->expenditureTools(),     // set_expenditure
        $this->campaignSaveTaxTools(), // capture_salary_sacrifice, capture_spouse_*, capture_pension_history,
        capture_charitable_giving
    );
}
```

Provider format switch (last block of `getTools`): if `Cache::get('ai_provider')` is `xai`, returns the tool list raw (already in OpenAI function-calling shape). Otherwise converts `parameters` → `input_schema` for the Anthropic Messages API.

Handoff tools (`delegate_to_capture`, `capture_complete`) are emitted only by `AiToolDefinitions::handoffTools()` and surfaced via `toolsListOverride` from `OnboardingChatDirector::handleInlineCapture`. They never appear in the normal `getTools()` list.

Onboarding extraction tools (`capture_personal_details`, `capture_spouse_details`, `capture_dependants`, `capture_work_details`) are emitted only by `AiToolDefinitions::onboardingExtractionTools()` and surfaced via `toolsListOverride` from `OnboardingChatDirector` during grouped-extract turns.

Advice-mode write-tool stripping

`AdviceFyn::WRITE_TOOLS` (defined at `app/Services/AI/AdviceFyn.php:151–175`) lists every tool stripped from the advice-mode catalogue:

```
private const WRITE_TOOLS = [
    'create_savings_account', 'create_investment_account', 'create_holding',
    'create_pension', 'create_property', 'create_mortgage',
    'create_protection_policy', 'create_asset', 'create_liability',
    'create_estate_gift', 'create_chattel', 'create_business_interest',
    'create_trust', 'create_family_member', 'create_will', 'update_will',
    'create_power_of_attorney', 'update_power_of_attorney',
    'update_record', 'delete_record', 'update_profile', 'set_expenditure',
    'capture_personal_details', 'capture_spouse_details',
    'capture_dependants', 'capture_work_details',
    'create_goal', 'create_life_event', 'create_what_if_scenario',
    'navigate_to_page',
];
```

Stripping happens in `AdviceFyn` (line ~534): `array_values(array_diff($names, self::WRITE_TOOLS))`.

Tool dispatch (`CoordinatingAgent::executeTool`)

`app/Agents/CoordinatingAgent.php:770–941` is the single dispatch point. Behaviour:

- 1. Normalise input — convert string "null" → null, decode HTML entities (xAI sometimes encodes & as &).
- 2. Compute `isPreviewUser` — `$user->is_preview_user && ! EvalBypassGate::isActive($user)`.
- 3. **Audit chain append** (`appendAuditEvent`) — forensic chain row, status `dispatched` (S0.12).
- 4. Emit eval `AgentDecision` event with name + args.
- 5. **Prerequisite gate check** (`PrerequisiteGateService::canExecuteTool`) — blocks if required data is missing; returns `['blocked' => true, 'reason' => ..., 'missing_data' => ..., 'suggested_action' => ..., 'instruction' => ...]`.
- 6. `match($toolName) → handler method`.
- 7. Audit chain completion row.
- 8. Catch `ValidationException`, `QueryException`, generic `Exception` → return `['error' => true, 'error_type' => ..., 'message' => ...]`.

Master tool list

#	Tool name	Mode	Preview-blocked?	Schema source	Handler in <code>CoordinatingAgent.php</code>
1	<code>navigate_to_page</code>	Onboarding only (advice strips)	No	<code>navigationTools()</code>	<code>handleNavigation :1062</code>
2	<code>list_records</code>	Both	No	<code>analysisTools()</code>	<code>handleListRecords :1398</code>
3	<code>list_goals</code>	Both	No	<code>analysisTools()</code>	<code>handleListGoals :1543</code>
4	<code>list_life_events</code>	Both	No	<code>analysisTools()</code>	<code>handleListLifeEvents :1582</code>
5	<code>get_module_analysis</code>	Both	No	<code>analysisTools()</code>	<code>handleModuleAnalysis :1620</code>
6	<code>get_recommendations</code>	Both	No	<code>analysisTools()</code>	<code>handleRecommendations :1744</code>
7	<code>search_conversation_index</code>	Both	No	<code>analysisTools()</code>	<code>handleSearchConversationIndex :167</code>
8	<code>get_tax_information</code>	Both	No	<code>taxTools()</code>	<code>handleTaxInformation :1847</code>
9	<code>generate_financial_plan</code>	Both	No	<code>planGenerationTools()</code>	<code>handleFinancialPlan :1890</code>
10	<code>get_subscription_status</code>	Both	No (read-only)	<code>billingTools()</code>	<code>handleGetSubscriptionStatus :1764</code>
11	<code>list_invoices</code>	Both	No	<code>billingTools()</code>	<code>handleListInvoices :1797</code>
12	<code>get_current_plan</code>	Both	No	<code>billingTools()</code>	<code>handleGetCurrentPlan :1818</code>
13	<code>create_what_if_scenario</code>	Onboarding only	Yes (no whatIfTools in preview)	<code>whatIfTools()</code>	<code>handleCreateWhatIfScenario :1723</code>

#	Tool name	Mode	Preview-blocked?	Schema source	Handler in <code>CoordinatingAgent.php</code>
14	<code>create_goal</code>	Onboarding only	Yes	<code>goalAndEventTools()</code>	<code>handleCreateGoal :1920</code>
15	<code>create_life_event</code>	Onboarding only	Yes	<code>goalAndEventTools()</code>	<code>handleCreateLifeEvent :1975</code>
16	<code>create_savings_account</code>	Onboarding only	Yes	<code>accountCreationTools()</code>	<code>handleCreateSavingsAccount :2023</code>
17	<code>create_investment_account</code>	Onboarding only	Yes	<code>accountCreationTools()</code>	<code>handleCreateInvestmentAccount :210</code>
18	<code>create_holding</code>	Onboarding only	Yes	<code>accountCreationTools()</code>	<code>handleCreateHolding :2243</code>
19	<code>create_pension</code>	Onboarding only	Yes	<code>accountCreationTools()</code>	<code>handleCreatePension :2325</code>
20	<code>create_property</code>	Onboarding only	Yes	<code>propertyCreationTools()</code>	<code>handleCreateProperty :2437</code>
21	<code>create_mortgage</code>	Onboarding only	Yes	<code>propertyCreationTools()</code>	<code>handleCreateMortgage :2568</code>
22	<code>create_protection_policy</code>	Onboarding only	Yes	<code>protectionCreationTools()</code>	<code>handleCreateProtectionPolicy :2659</code>
23	<code>create_asset</code>	Onboarding only	Yes	<code>estateCreationTools()</code>	<code>handleCreateEstateAsset :2876</code>
24	<code>create_liability</code>	Onboarding only	Yes	<code>estateCreationTools()</code>	<code>handleCreateEstateLiability :2923</code>
25	<code>create_estate_gift</code>	Onboarding only	Yes	<code>estateCreationTools()</code>	<code>handleCreateEstateGift :2985</code>
26	<code>create_will</code>	Onboarding only	Yes	<code>estateCreationTools()</code>	<code>handleCreateWill :3032</code>
27	<code>update_will</code>	Onboarding only	Yes	<code>estateCreationTools()</code>	<code>handleUpdateWill :3077</code>
28	<code>create_power_of_attorney</code>	Onboarding only	Yes	<code>estateCreationTools()</code>	<code>handleCreatePowerOfAttorney :3136</code>
29	<code>update_power_of_attorney</code>	Onboarding only	Yes	<code>estateCreationTools()</code>	<code>handleUpdatePowerOfAttorney :3199</code>
30	<code>create_family_member</code>	Onboarding only	Yes	<code>additionalCreationTools()</code>	<code>handleCreateFamilyMember :3539</code>
31	<code>create_trust</code>	Onboarding only	Yes	<code>additionalCreationTools()</code>	<code>handleCreateTrust :3645</code>
32	<code>create_business_interest</code>	Onboarding only	Yes	<code>additionalCreationTools()</code>	<code>handleCreateBusinessInterest :3721</code>
33	<code>create_chattel</code>	Onboarding only	Yes	<code>additionalCreationTools()</code>	<code>handleCreateChattel :3781</code>
34	<code>update_record</code>	Onboarding only	Yes	<code>dataModificationTools()</code>	<code>handleUpdateRecord :4092</code>
35	<code>delete_record</code>	Onboarding only	Yes	<code>dataModificationTools()</code>	<code>handleDeleteRecord :4200</code>
36	<code>update_profile</code>	Onboarding only	Yes	<code>profileTools()</code>	<code>handleUpdateProfile :4301</code>
37	<code>set_expenditure</code>	Onboarding only	Yes	<code>expenditureTools()</code>	<code>handleSetExpenditure :3840</code>
38	<code>capture_salary_sacrifice</code>	Onboarding (campaign)	Yes	<code>campaignSaveTaxTools()</code>	<code>handleCaptureSalarySacrifice :3905</code>
39	<code>capture_spouse_work_status</code>	Onboarding (campaign)	Yes	<code>campaignSaveTaxTools()</code>	<code>handleCaptureSpouseWorkStatus :393</code>
40	<code>capture_spouse_household_data</code>	Onboarding (campaign)	Yes	<code>campaignSaveTaxTools()</code>	<code>handleCaptureSpouseHouseholdData :3966</code>

#	Tool name	Mode	Preview-blocked?	Schema source	Handler in <code>CoordinatingAgent.php</code>
41	<code>capture_spouse_non_working_assets</code>	Onboarding (campaign)	Yes	<code>campaignSaveTaxTools()</code>	<code>handleCaptureSpouseNonWorkingAssets:3995</code>
42	<code>capture_pension_history</code>	Onboarding (campaign)	Yes	<code>campaignSaveTaxTools()</code>	<code>handleCapturePensionHistory:4022</code>
43	<code>capture_charitable_giving</code>	Onboarding (campaign)	Yes	<code>campaignSaveTaxTools()</code>	<code>handleCaptureCharitableGiving:406</code>
44	<code>capture_personal_details</code>	Onboarding grouped-extract only	n/a	<code>onboardingExtractionTools()</code> (override)	<code>handleCapturePersonalDetails:1074</code>
45	<code>capture_spouse_details</code>	Onboarding grouped-extract only	n/a	<code>onboardingExtractionTools()</code> (override)	<code>handleCaptureSpouseDetails:1183</code>
46	<code>capture_dependants</code>	Onboarding grouped-extract only	n/a	<code>onboardingExtractionTools()</code> (override)	<code>handleCaptureDependants:1257</code>
47	<code>capture_work_details</code>	Onboarding grouped-extract only	n/a	<code>onboardingExtractionTools()</code> (override)	<code>handleCaptureWorkDetails:1325</code>
48	<code>delegate_to_capture</code>	Advice-mode handoff stub	n/a	<code>handoffTools()</code> (override)	inline (no handler — returns <code>['action' => 'handoff', ...]</code>)
49	<code>capture_complete</code>	Onboarding handoff stub	n/a	<code>handoffTools()</code> (override)	inline (no handler — returns <code>['action' => 'handoff', ...]</code>)

Per-tool source files

Full schema + handler code for every tool, grouped by category:

- `01-read-tools.md` — Read-only tools available in both modes (12 tools)
- `02-creation-financial.md` — Account/property/protection creates (8 tools)
- `03-creation-estate.md` — Estate creates: gift, will, LPA, asset, liability, trust, chattel, business, family member (10 tools)
- `04-creation-goals-whatif.md` — Goal, life event, what-if scenario (3 tools)
- `05-mutation.md` — `update_record`, `delete_record`, `update_profile`, `set_expenditure` (4 tools)
- `06-onboarding-capture.md` — Grouped-extract tools (4 tools)
- `07-savetax-campaign.md` — SaveTax campaign capture tools (6 tools)
- `08-handoff.md` — `delegate_to_capture`, `capture_complete` (2 stubs)
- `09-shared-helpers.md` — Shared helper methods called by handlers: `previewBlocked`, `validateToolInput`, `checkForDuplicate`, `resolveModel`, `resolveFamilyNames`, `summariseToolAnalysis`, `resolvePropertyId`, `invalidateModuleCache`, `educationStatusForAge`, plus the `executeTool` dispatcher itself.

Special routing notes

- `update_profile` with `section: expenditure` internally redirects to `handleSetExpenditure` (`CoordinatingAgent.php:4310-4312`) — the LLM can use either tool, the result is the same row write.
- `update_will` when no will exists falls through to `handleCreateWill` (`CoordinatingAgent.php:3097-3099`) — create-on-update semantics for the singleton will record.
- **In-turn idempotency** — `handleCreateProtectionPolicy` (life, CI, IP branches) checks for an existing row matching `policy_type` + `sum_assured` + `provider` created within the last 60 seconds and returns `['duplicate' => true, ...]` instead of creating a second row. This is BS-17 — grok-4-1-fast occasionally emits the same create twice in one multi-entity message.
- **delete_record two-phase confirmation** — first call returns `['requires_confirmation' => true, 'confirmation_token' => sha256(...)]`; second call must echo the exact token. Token is bound to `(user_id, entity_type, entity_id, today's date)` so it cannot be replayed across days.
- **update_record allowlist** — fields are validated against `App\Constants\UpdateRecordAllowlist::MAP[$entityType]`. `update_record_schema()` builds a `oneOf` JSON Schema branch per `entity_type` so the LLM cannot invent field names. The runtime handler re-checks the allowlist (defence-in-depth).